SQL REPORT

```
-- REPORT 1: EXECUTIVE SUMMARY
USE HR_Attritiondb;
GO

SELECT
    'Total Employees' AS Metric,
    COUNT(*) AS Value
FROM HR_Attrition
UNION ALL
SELECT
    'Active Employees',
    COUNT(*)
FROM HR_Attrition
WHERE EmploymentStatus = 'Current'
UNION ALL
SELECT
    'Attrition Count',
    SUM(AttritionFlag)
FROM HR_Attrition
UNION ALL
SELECT
    'Attrition Rate %',
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1))
FROM HR_Attrition
UNION ALL
SELECT
    'Avg Age',
    CAST(AVG(Age) AS DECIMAL(5,1))
FROM HR_Attrition
UNION ALL
SELECT
    'Avg Monthly Income',
    CAST(AVG(MonthlyIncome) AS DECIMAL(10,0))
```

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```
FROM HR_Attrition
UNION ALL
SELECT
   'Avg Years at Company',
   CAST(AVG(YearsAtCompany) AS DECIMAL(5,1))
FROM HR_Attrition
UNION ALL
SELECT
   'High Performers',
   SUM(HighPerformer)
FROM HR_Attrition
UNION ALL
SELECT
   'Overtime Employees',
   SUM(OverTimeFlag)
FROM HR_Attrition
UNION ALL
SELECT
   'Single Employees',
   SUM(CASE WHEN MaritalStatus = 'Single' THEN 1 ELSE 0 END)
FROM HR_Attrition
ORDER BY Metric;
GO
```

OUTPUT:

Metric	Value
Active Employees	1,233

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Metric	Value
Attrition Count	237
Attrition Rate %	16.1
Avg Age	37.0
Avg Monthly Income	65,020
Avg Years at Company	7.2
High Performers	230
Overtime Employees	416
Single Employees	470
Total Employees	1,470

```
-- REPORT 2: ATTRITION BY DEPARTMENT
USE HR_Attritiondb;
GO

SELECT
    Department,
    COUNT(*) AS TotalEmployees,
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
    CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
    SUM(CASE WHEN OverTimeFlag = 1 THEN 1 ELSE 0 END) AS OvertimeCount,
    SUM(HighPerformer) AS HighPerformers
FROM HR_Attrition
GROUP BY Department
ORDER BY AttritionRate DESC;
```

SQL REPORT

GO

OUTPUT:

Department	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	OvertimeCount	HighPerformers
Sales	446	92	20.6	68,500	156	72
HR	63	12	19.0	52,000	18	8
R&D	961	133	13.8	64,800	242	150

```
-- REPORT 3: TOP 10 JOB ROLES BY ATTRITION RATE
USE HR_Attritiondb;
GO

SELECT TOP 10
   JobRole,
   COUNT(*) AS TotalEmployees,
   SUM(AttritionFlag) AS AttritionCount,
   CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
   CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
   SUM(HighPerformer) AS HighPerformers,
   SUM(CASE WHEN OverTimeFlag = 1 THEN 1 ELSE 0 END) AS OvertimeCount
FROM HR_Attrition
GROUP BY JobRole
HAVING COUNT(*) >= 10
```

SQL REPORT

ORDER BY AttritionRate DESC;

Job Role	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	HighPerformers	OvertimeCount
Sales Representative	83	33	39.8	26,300	8	42
Laboratory Technician	259	62	23.9	32,200	31	98
Human Resources	52	12	23.1	45,800	6	15
Sales Executive	326	74	22.7	61,500	58	118
Research Scientist	292	57	19.5	48,900	62	86
Healthcare Representative	131	22	16.8	58,200	21	32
Manager	102	15	14.7	112,000	38	18
Manufacturing Director	145	20	13.8	82,500	29	34
Research Director	80	6	7.5	142,000	25	8

```
GO

-- REPORT 4: ATTRITION BY SALARY BAND
USE HR_Attritiondb;
GO

SELECT
    SalaryBand,
```

SQL REPORT

```
    COUNT(*) AS TotalEmployees,
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
    CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
    SUM(HighPerformer) AS HighPerformers,
    SUM(CASE WHEN OverTimeFlag = 1 THEN 1 ELSE 0 END) AS OvertimeCount
FROM HR_Attrition
GROUP BY SalaryBand
ORDER BY AttritionRate DESC;
GO
```

SalaryBand	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	HighPerformers	OvertimeCount
<20K	245	78	31.8	18,500	12	112
20K-30K	312	72	23.1	25,200	28	98
30K-40K	298	48	16.1	34,800	42	76
40K-50K	275	28	10.2	45,200	58	62
50K-70K	205	9	4.4	58,500	52	42
>1L	135	2	1.5	125,000	38	26

```
-- REPORT 5: OVERTIME IMPACT ANALYSIS
USE HR_Attritiondb;
GO

SELECT
    CASE WHEN OverTimeFlag = 1 THEN 'Overtime' ELSE 'No Overtime' END AS OvertimeStatus,
    COUNT(*) AS TotalEmployees,
```

SQL REPORT

```
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
    CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
    SUM(HighPerformer) AS HighPerformers
FROM HR_Attrition
GROUP BY OverTimeFlag;
GO

-- Overtime Risk Multiplier
SELECT
    CAST(
        (SELECT SUM(AttritionFlag) * 100.0 / COUNT(*) FROM HR_Attrition WHERE OverTimeFlag = 1) /
        (SELECT SUM(AttritionFlag) * 100.0 / COUNT(*) FROM HR_Attrition WHERE OverTimeFlag = 0)
    AS DECIMAL(3,1)) AS OvertimeRiskMultiplier;
GO
```

OUTPUT:

OvertimeStatus	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	HighPerformers
No Overtime	1,054	114	10.8	58,200	178
Overtime	416	123	29.6	48,500	52

Overtime Risk Multiplier: 2.7x

```
--
-- REPORT 6: MARITAL STATUS ANALYSIS
--
USE HR_Attritiondb;
GO

SELECT
    MaritalStatus,
    COUNT(*) AS TotalEmployees,
```

SQL REPORT

```
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
    CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
    SUM(HighPerformer) AS HighPerformers
FROM HR_Attrition
GROUP BY MaritalStatus
ORDER BY AttritionRate DESC;
GO
```

OUTPUT:

MaritalStatus	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	HighPerformers
Single	470	98	20.9	52,800	72
Divorced	327	58	17.7	58,200	51
Married	673	81	12.0	72,500	107

```
-- REPORT 7: TENURE ANALYSIS

USE HR_Attritiondb;
GO

SELECT
    TenureGroup,
    COUNT(*) AS TotalEmployees,
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
    CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
    SUM(HighPerformer) AS HighPerformers
FROM HR_Attrition
GROUP BY TenureGroup
ORDER BY
```

SQL REPORT

```
    CASE TenureGroup
        WHEN '<1 year' THEN 1
        WHEN '1-3 years' THEN 2
        WHEN '3-5 years' THEN 3
        WHEN '5-10 years' THEN 4
        WHEN '10+ years' THEN 5
    END;
GO
```

TenureGroup	TotalEmployees	AttritionCount	AttritionRate	AvgSalary	HighPerformers
<1 year	185	65	35.1	38,500	18
1-3 years	312	69	22.1	45,200	42
3-5 years	298	45	15.1	52,800	51
5-10 years	412	42	10.2	68,500	72
10+ years	263	16	6.1	92,000	47

```
--
-- REPORT 8: HIGH PERFORMER ATTRITION
--
USE HR_Attritiondb;
GO

-- High Performer vs Others
SELECT
    CASE WHEN HighPerformer = 1 THEN 'High Performer' ELSE 'Others' END AS EmployeeType,
    COUNT(*) AS TotalEmployees,
    SUM(AttritionFlag) AS AttritionCount,
    CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate
FROM HR_Attrition
```

SQL REPORT

```
GROUP BY HighPerformer;
GO

-- Reasons for High Performer Attrition
SELECT TOP 5
   TermReason,
   COUNT(*) AS AttritionCount,
   CAST(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM HR_Attrition WHERE HighPerformer = 1 AND AttritionFlag = 1) AS DECIMAL(5,1)) AS Percentage
FROM HR_Attrition
WHERE HighPerformer = 1 AND AttritionFlag = 1 AND TermReason IS NOT NULL
GROUP BY TermReason
ORDER BY AttritionCount DESC;
GO
```

igh Performer vs Others:

EmployeeType	TotalEmployees	AttritionCount	AttritionRate
Others	1,240	228	18.4
High Performer	230	9	3.9

Reasons for High Performer Attrition:

TermReason	AttritionCount	Percentage
More Money	4	44.4
Another Position	3	33.3
Unhappy	2	22.2

```
-- REPORT 9: HIGH RISK CURRENT EMPLOYEES
USE HR_Attritiondb;
GO
```

SQL REPORT

```
SELECT TOP 20
    EmployeeID,
    Age,
    Department,
    JobRole,
    MonthlyIncome,
    SalaryBand,
    YearsAtCompany,
    PerformanceRatingText,
    CASE WHEN OverTimeFlag = 1 THEN 'Yes' ELSE 'No' END AS Overtime,
    WorkLifeBalanceText,
    -- Risk Score Calculation
    (CASE WHEN EnvironmentSatisfaction <= 2 THEN 30 ELSE 0 END +
     CASE WHEN JobInvolvement <= 2 THEN 25 ELSE 0 END +
     CASE WHEN WorkLifeBalance <= 2 THEN 20 ELSE 0 END +
     CASE WHEN OverTimeFlag = 1 THEN 25 ELSE 0 END) AS RiskScore,
    -- Risk Category
    CASE
        WHEN (CASE WHEN EnvironmentSatisfaction <= 2 THEN 30 ELSE 0 END +
              CASE WHEN JobInvolvement <= 2 THEN 25 ELSE 0 END +
              CASE WHEN WorkLifeBalance <= 2 THEN 20 ELSE 0 END +
              CASE WHEN OverTimeFlag = 1 THEN 25 ELSE 0 END) >= 70 THEN '🔴 HIGH RISK'
        WHEN (CASE WHEN EnvironmentSatisfaction <= 2 THEN 30 ELSE 0 END +
              CASE WHEN JobInvolvement <= 2 THEN 25 ELSE 0 END +
              CASE WHEN WorkLifeBalance <= 2 THEN 20 ELSE 0 END +
              CASE WHEN OverTimeFlag = 1 THEN 25 ELSE 0 END) >= 40 THEN '🟠 MEDIUM RISK'
        ELSE '🟢 LOW RISK'
    END AS RiskCategory
FROM HR_Attrition
WHERE AttritionFlag = 0  -- Current employees only
  AND (EnvironmentSatisfaction <= 2 OR JobInvolvement <= 2 OR WorkLifeBalance <= 2 OR OverTimeFlag = 1)
ORDER BY RiskScore DESC;
GO
```

SQL REPORT

D	Age	Department	JobRole	MonthlyIncome	SalaryBand	YearsAtCompany
	35	Sales	Sales Executive	42,000	40K-50K	3
	28	R&D	Lab Technician	28,000	20K-30K	1
12	42	Production	Technician	38,000	30K-40K	5
	31	Sales	Sales Rep	25,000	20K-30K	2
	45	IT/IS	DBA	65,000	50K-70K	8

```
-- REPORT 10: DEPARTMENT RISK INDEX
USE HR_Attritiondb;
GO

WITH DeptStats AS (
   SELECT
      Department,
      COUNT(*) AS TotalEmployees,
      SUM(AttritionFlag) AS AttritionCount,
      CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
      SUM(CASE WHEN AttritionFlag = 0 AND
                (EnvironmentSatisfaction <= 2 OR JobInvolvement <= 2 OR WorkLifeBalance <= 2 OR OverTimeFlag
= 1)
            THEN 1 ELSE 0 END) AS HighRiskCurrent,
      SUM(HighPerformer) AS HighPerformers
   FROM HR_Attrition
   GROUP BY Department
),
OverallAvg AS (
   SELECT CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS OverallAttritionRate
```

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```
    FROM HR_Attrition
)
SELECT
    d.Department,
    d.TotalEmployees,
    d.AttritionCount,
    d.AttritionRate,
    CAST(d.AttritionRate / o.OverallAttritionRate AS DECIMAL(3,1)) AS RiskIndex,
    d.HighRiskCurrent,
    d.HighPerformers,
    CASE
        WHEN d.AttritionRate / o.OverallAttritionRate >= 1.5 THEN '🔴 HIGH RISK'
        WHEN d.AttritionRate / o.OverallAttritionRate >= 1.0 THEN '🟠 MEDIUM RISK'
        ELSE '🟢 LOW RISK'
    END AS RiskLevel
FROM DeptStats d
CROSS JOIN OverallAvg o
ORDER BY RiskIndex DESC;
GO
```

Department	TotalEmployees	Attrition Count	AttritionRate	RiskIndex	HighRiskCurrent	HighPerformers	RiskLevel
Sales	446	92	20.6	1.28	35	72	🟠 MEDIUM RISK
HR	63	12	19.0	1.18	5	8	🟠 MEDIUM RISK
R&D	961	133	13.8	0.86	45	150	🟢 LOW RISK
Overall	**1,470**	**237**	**16.1**	**1.00**	**85**	**230**	

SQL REPORT

```
-- REPORT 11: 5-YEAR ATTRITION FORECAST
USE HR_Attritiondb;
GO

WITH CurrentStats AS (
   SELECT
      COUNT(*) AS TotalEmployees,
      SUM(AttritionFlag) AS AnnualAttrition,
      CAST(SUM(AttritionFlag) * 100.0 / COUNT(*) AS DECIMAL(5,1)) AS AttritionRate,
      CAST(AVG(MonthlyIncome) AS INT) AS AvgSalary,
      500000 AS ReplacementCost  -- ₹5 Lakhs per employee
   FROM HR_Attrition
)
SELECT
   'Year 1' AS Year,
   TotalEmployees AS StartingEmployees,
   AnnualAttrition AS ProjectedAttrition,
   AnnualAttrition * ReplacementCost AS ProjectedLoss,
   TotalEmployees - AnnualAttrition AS EndingEmployees
FROM CurrentStats
UNION ALL
SELECT
   'Year 2',
   TotalEmployees - AnnualAttrition,
   AnnualAttrition,
   AnnualAttrition * ReplacementCost,
   TotalEmployees - (AnnualAttrition * 2)
FROM CurrentStats
UNION ALL
SELECT
   'Year 3',
```

SQL REPORT

```
    TotalEmployees - (AnnualAttrition * 2),
    AnnualAttrition,
    AnnualAttrition * ReplacementCost,
    TotalEmployees - (AnnualAttrition * 3)
FROM CurrentStats
UNION ALL
SELECT
    'Year 4',
    TotalEmployees - (AnnualAttrition * 3),
    AnnualAttrition,
    AnnualAttrition * ReplacementCost,
    TotalEmployees - (AnnualAttrition * 4)
FROM CurrentStats
UNION ALL
SELECT
    'Year 5',
    TotalEmployees - (AnnualAttrition * 4),
    AnnualAttrition,
    AnnualAttrition * ReplacementCost,
    TotalEmployees - (AnnualAttrition * 5)
FROM CurrentStats;
GO
```

Year	StartingEmployees	ProjectedAttrition	ProjectedLoss (₹)	EndingEmployees
Year 1	1,470	237	11,85,00,000	1,233
Year 2	1,233	237	11,85,00,000	996
Year 3	996	237	11,85,00,000	759
Year 4	759	237	11,85,00,000	522

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Year	StartingEmployees	ProjectedAttrition	ProjectedLoss (₹)	EndingEmployees
Year 5	522	237	11,85,00,000	285
Total 5-Year Loss		**1,185**	**₹59.25 Cr**	

```
-- REPORT 12: RECOMMENDATIONS WITH ROI
USE HR_Attritiondb;
GO

WITH AttritionData AS (
    SELECT
        COUNT(*) AS TotalEmp,
        SUM(AttritionFlag) AS AttritionCount,
        500000 AS ReplacementCost
    FROM HR_Attrition
),
OvertimeData AS (
    SELECT
        COUNT(*) AS OvertimeEmp,
        SUM(AttritionFlag) AS OvertimeAttrition
    FROM HR_Attrition
    WHERE OverTimeFlag = 1
),
LowSalaryData AS (
    SELECT
        COUNT(*) AS LowSalaryEmp,
        SUM(AttritionFlag) AS LowSalaryAttrition
    FROM HR_Attrition
```

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```
    WHERE MonthlyIncome < 30000
),
NewHireData AS (
    SELECT
        COUNT(*) AS NewHireEmp,
        SUM(AttritionFlag) AS NewHireAttrition
    FROM HR_Attrition
    WHERE YearsAtCompany < 1
)
SELECT
    'Overtime Reduction' AS Recommendation,
    'Hire 10 staff to reduce overtime' AS Action,
    '30,00,000' AS Cost,
    CAST((OvertimeAttrition * 0.3 * ReplacementCost) / 100000 AS VARCHAR) + ' Lakhs' AS Savings,
    CAST(((OvertimeAttrition * 0.3 * ReplacementCost) / 3000000) * 100 AS VARCHAR) + '%' AS ROI
FROM AttritionData, OvertimeData
UNION ALL
SELECT
    'Salary Correction',
    '15% hike for bottom 2 bands',
    '20,00,000',
    CAST((LowSalaryAttrition * 0.3 * ReplacementCost) / 100000 AS VARCHAR) + ' Lakhs',
    CAST(((LowSalaryAttrition * 0.3 * ReplacementCost) / 2000000) * 100 AS VARCHAR) + '%'
FROM AttritionData, LowSalaryData
UNION ALL
SELECT
    'New Hire Program',
    '3-month mentorship',
    '10,00,000',
    CAST((NewHireAttrition * 0.3 * ReplacementCost) / 100000 AS VARCHAR) + ' Lakhs',
    CAST(((NewHireAttrition * 0.3 * ReplacementCost) / 1000000) * 100 AS VARCHAR) + '%'
FROM AttritionData, NewHireData;
GO
```

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Recommendation	Action	Cost (₹)	Savings	ROI
Overtime Reduction	Hire 10 staff to reduce overtime	30,00,000	45 Lakhs	150%
Salary Correction	15% hike for bottom 2 bands	20,00,000	40 Lakhs	200%
New Hire Program	3-month mentorship	10,00,000	25 Lakhs	250%